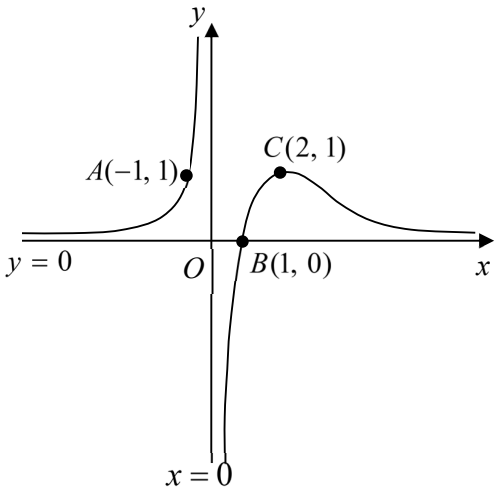
RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Additional Practice Questions for Chapter 8B: Curve Transformations

Transformations

1



The diagram shows the graph of $y = \mathrm{f}(x)$. The curve passes through the points $A(-1,1)$ and $B(1,0)$, and has a maximum point at $C(2,1)$. The curve has asymptotes $x=0$ and $y=0$. Sketch, on separate diagrams, the graphs of

(i) $y = \mathrm{f}(2x-1)$, **(ii)** $y = \mathrm{f}(-x-1)$, **(iii)** $y = 2\mathrm{f}(2x)+1$.

In each case, state the coordinates of the points corresponding to A, B, and C, and the equations of the asymptotes, whenever applicable. **[9]**

2 State precisely how each of the following curves may be obtained by applying a sequence of geometrical transformations from the given starting curve.

(a) $y = 3\mathrm{e}^{x+2}$ (starting curve : $y = \mathrm{e}^{x}$),

(b) $y = 1 - 2\sqrt{1-x}$ (starting curve : $y = \sqrt{x}$). **[7]**

3 The transformations A, B and C are as follows:

A: A translation of 1 unit in the negative y-direction;

B: A stretching parallel to the y-axis (with x-axis invariant) by a factor of 2;

C: A reflection about the x-axis.

A curve undergoes in succession, the transformations A, B and C, and the equation of the resulting curve is $y = -2x^2 - 6$.

(i) Determine the equation of the curve before the transformations were effected. **[3]**

Suppose the curve undergoes in succession, the above transformations in the reverse order (i.e. C, B and A).

(ii) Determine the equation of the resulting curve. **[3]**

(iii) Suggest one amendment to the above transformations for the reverse order such that the resulting equation is still $y = -2x^2 - 6$. **[1]**

4 9758/2021/01/Q6

A curve C has equation $y = \dfrac{1}{\sqrt{4ax - x^2}}$, where $a > 0$.

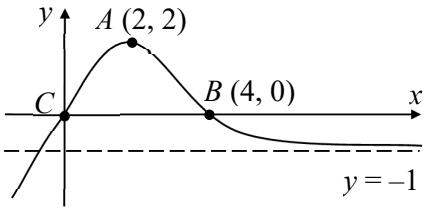
(a) Sketch C and give the equations of any asymptotes, in terms of a where appropriate. **[4]**

(b) Find the smallest possible value of y in terms of a. **[1]**

(c) Describe the transformation that maps the graph of C onto the graph of $y = \dfrac{1}{\sqrt{4a^2 - x^2}}$. **[3]**

Transformations involving Modulus Functions

5 The graph of $y = \mathrm{f}(2x) - 1$ is shown.



Sketch on separate diagrams, the graphs of

(i) $y = -2\,\mathrm{f}(2x)$,

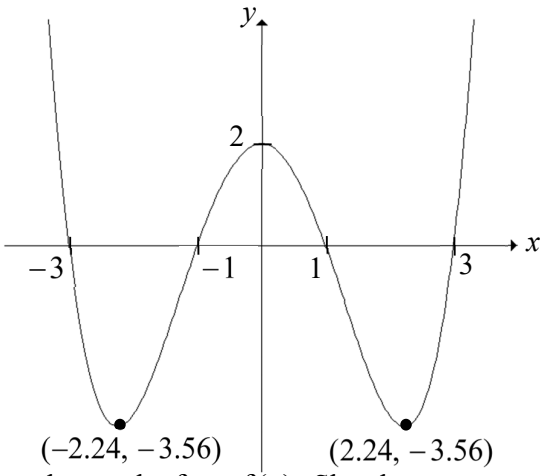
(ii) $y = \mathrm{f}(x)$,

(iii) $y = \mathrm{f}(|x|)$.

Your sketch should show clearly the coordinates corresponding to A, B and C, and the equations of asymptotes (if any).

[8]

6

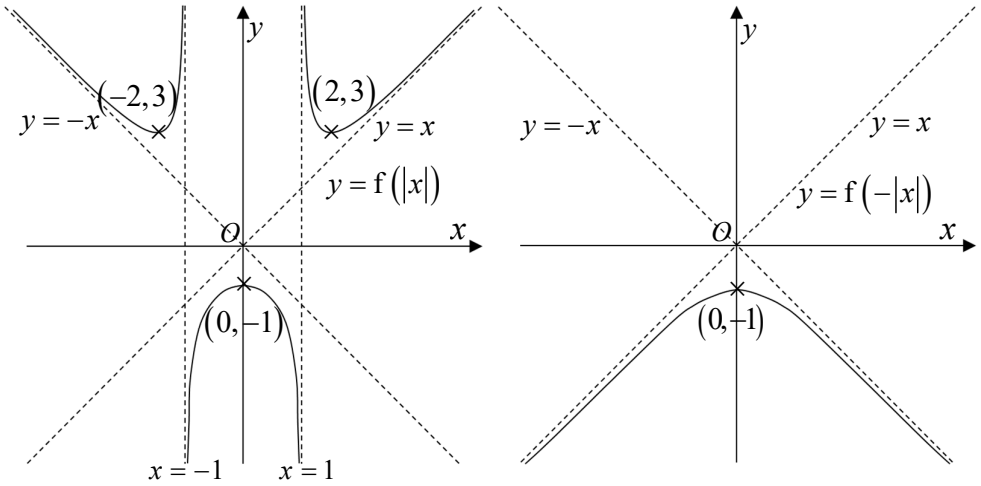


The diagram shows the graph of $y = \mathrm{f}(x)$. Sketch, on separate diagrams, the graphs of

(i) $y = -\mathrm{f}(x)$, **(ii)** $y = |\mathrm{f}(x)|$, **(iii)** $y = \mathrm{f}(|x| - 1)$, **(iv)** $y = \mathrm{f}(1 - |2x|)$. **[12]**

7 MI Promo 9758/2018/PU2/02/Q4a

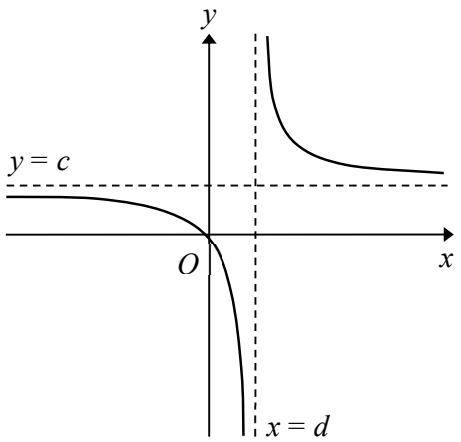
The graphs of $y = \mathrm{f}(|x|)$ and $y = \mathrm{f}(-|x|)$ are shown below.



Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of any turning points and intersections with the axes. [2]

8 VJC Mid-yr 9740/2006/Q11 Modified

The following shows the graph of $y = \dfrac{ax}{bx-1}$.



Sketch, on separate diagrams, the graphs of

(a) $y = \dfrac{a|x|}{b|x|-1}$, **(b)** $y = \left|\dfrac{ax}{bx-1}\right|$, **(c)** $y = \left|\dfrac{ax}{bx+1}\right|$.

Given that $c = d = \dfrac{1}{2}$, write down the values of a and b. **[8]**

Transformations involving Conic Sections

9 VJC Prelim 9740/2009/ 01/Q6

A curve C_1 has equation $x^2 - a^2y^2 = a^2$ where $a > 1$. Sketch C_1, indicating the axial intercepts, asymptotes and stationary points, if any. **[2]**

C_1 undergoes a single transformation to become C_2. Given that C_2 has a line of symmetry $y = 4$ and the point (4, 3) lies on C_2, find the value of a. **[3]**

10 MI Promo 9758/2018/PU1/Q8

The curve C_1 has equation $9x^2 + 18x + 4y^2 - 8y = 23$.

(i) By completing the square, show that the equation of C_1 can be expressed as $\frac{(x+1)^2}{2^2} + \frac{(y-1)^2}{3^2} = 1$. [2]

(ii) Sketch C_1, stating the coordinates of any points of intersection with the axes. [3]

(iii) By adding a suitable graph to part **(ii)**, solve the equation $\frac{(x+1)^2}{2^2} + \frac{x^2}{3^2} = 1$. [2]

(iv) State a sequence of transformations which transform the graph of C_1 to the graph of C_2 with equation $x^2 + y^2 = 1$. [3]

11 ACJC Prelim 9740/2011/02/Q5

The curves C_1 and C_2 have equations $9y^2 = (x+k)^2 - 9$ and $\frac{x^2}{k^2} + y^2 = 1$ respectively, where k is a real constant such that $k > 3$.

(a) Describe a sequence of transformations which transforms the graph of $x^2 - y^2 = 1$ to the graph of C_1. **[2]**

(b) **(i)** On the same diagram, sketch the graphs of C_1 and C_2, stating clearly the coordinates of any points of intersection with the axes and the equations of any asymptotes. **[5]**

(ii) Find the **range** of values of the positive constant a such that the equation $\frac{x^2}{a^2} + \frac{(x+k)^2 - 9}{9} = 1$ has four real roots. **[2]**

12 HCI Prelim 9740/2010/01/Q12 modified

The curves C_1 and C_2 have equations $(x-2)^2 = a^2(1-y^2)$ and $y = \dfrac{x^2-4}{x+1}$ respectively, where $1 < a < 2$.

State a sequence of transformations which transforms the graph of $x^2 + y^2 = 1$ to the graph of C_1 and describe the geometrical shape of C_1. **[4]**

(i) Sketch C_1 and C_2 on the same diagram, stating the coordinates of any points of intersection with the axes and the equations of any asymptotes. **[6]**

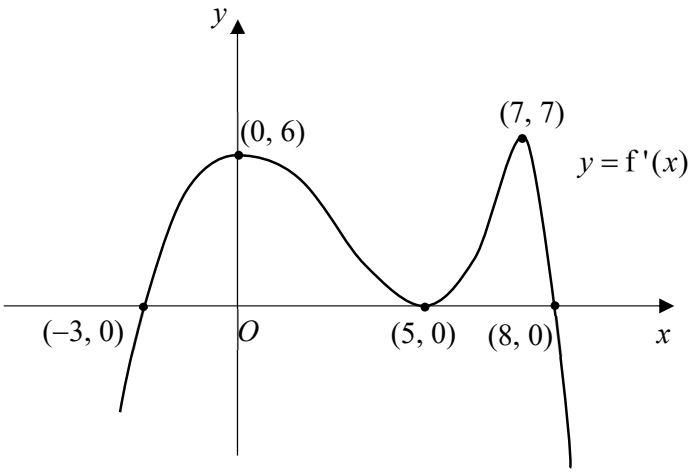
(ii) Show algebraically that the x-coordinates of the points of intersection of C_1 and C_2 satisfy the equation $(x+1)^2(x-2)^2 = a^2(x+1)^2 - a^2(x^2-4)^2$. **[2]**

(iii) Deduce the number of real roots of the equation in part **(ii)**. **[1]**

Graphs of Derivative Functions & Reciprocal Functions

13 DHS Prelim 9758/2018/02/Q4 modified

The diagram below shows the curve $y = \mathrm{f}'(x)$. It has turning points at (0, 6), (5, 0) and (7, 7) and intersects the x-axis at (–3, 0) and (8, 0).



(i) State the x-coordinates of the stationary points for the curve $y = \mathrm{f}(x)$. Hence determine the nature of these stationary points. [3]

(ii) State the range of values of x such that the curve $y = \mathrm{f}(x)$ is concave downwards. [2]

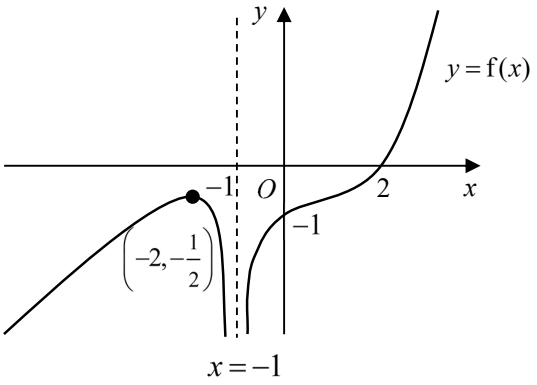
(iii) A student makes a claim on the following statement:

If the curve $y = \mathrm{f}(x)$ passes through the point (–3, 7), then $\mathrm{f}(x) > 7$ for $-3 < x < 5$.

Explain briefly whether the student is right to make such a claim. [1]

14 DHS Prelim 9740/2009/02/Q1 modified

The diagram below shows the graph of $y = \mathrm{f}(x)$. The curve has a maximum point at $\left(-2, -\frac{1}{2}\right)$ and an asymptote $x = -1$.



Sketch on separate diagrams, the graphs of

(i) $y = \mathrm{f}(2-x)$, **[2]**

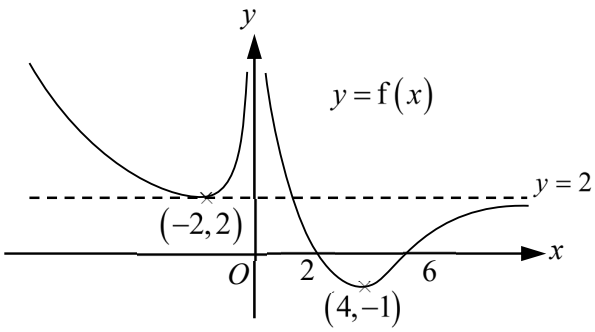
(ii) $y = \dfrac{1}{\mathrm{f}(x)}$, **[3]**

(iii) $y = \mathrm{f}'(x)$, **[3]**

showing, in each case, the coordinates of the axial intercepts, stationary points and asymptotes if they can be determined.

15 DHS Promo 9740/2015/01/Q4

The diagram below shows the graph of $y = \mathrm{f}(x)$ with asymptotes $y = 2$ and $x = 0$.
The curve has turning points at $(-2, 2)$ and $(4, -1)$.



On separate diagrams, sketch the graph of

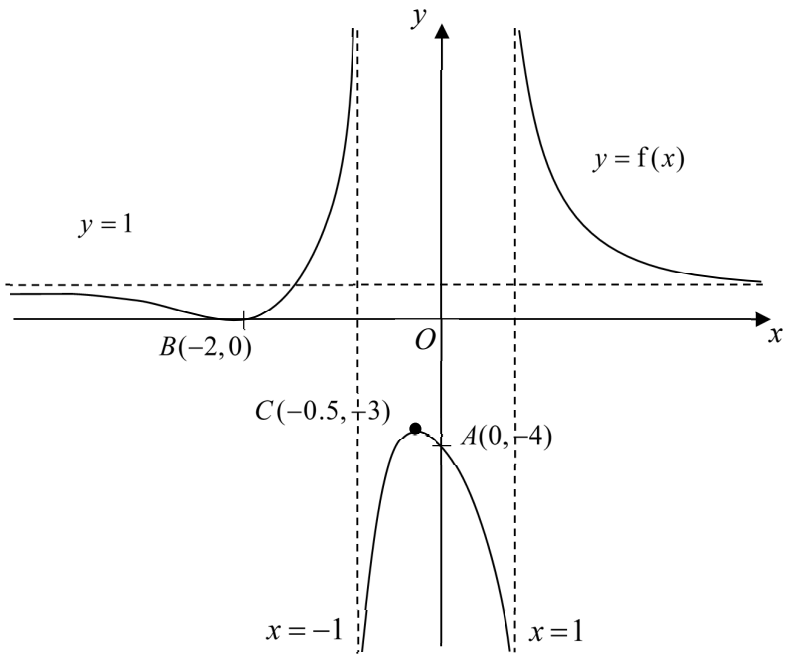
(i) $y = \dfrac{1}{\mathrm{f}(x)}$ and [3]

(ii) $y = \mathrm{f}(|x|)$ [2]

By adding a suitable graph to the graph in part **(ii)**, find the number of distinct roots of the equation $x^2 + \left[\mathrm{f}(|x|)\right]^2 = 4$. [2]

16 DHS Prelim 9758/2020/01/Q5

The diagram below shows the graph of $y = \mathrm{f}(x)$. It cuts the y–axis at $A(0,-4)$, has a minimum point at $B(-2,0)$ and a maximum point at $C(-0.5,-3)$. The equations of the horizontal asymptote is $y = 1$ and vertical asymptotes are $x = -1$ and $x = 1$ respectively. Moreover, when $x = 0$, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -3$.



Sketch, on separate diagrams, the graphs of

(i) $y = \mathrm{f}\left(|x-1|\right)$, [3]

(ii) $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

(iii) $y = \mathrm{f}'(x)$, [3]

stating clearly the equations of all the asymptotes, coordinates of any points of intersection with both axes and the coordinates of the points corresponding to A, B and C (if any and when possible).

Questions involving Functions

17 9758/2017/02/Q3

The curve $y = \mathrm{f}(x)$ cuts the axes at $(a, 0)$ and $(0, b)$. It is given that $\mathrm{f}^{-1}(x)$ exists. State, if it is possible to do so, the coordinates of the points where the following curves cut the axes.

(i) $y = \mathrm{f}(2x)$

(ii) $y = \mathrm{f}(x-1)$

(iii) $y = \mathrm{f}(2x-1)$

(iv) $y = \mathrm{f}^{-1}(x)$ [4]

18 9740/2012/01/Q7

A function f is said to be self-inverse if $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all x in the domain of f.

The function g is defined by $\mathrm{g}: x \to \dfrac{x+k}{x-1}$, $x \in \mathbb{R}, x \neq 1$, where k is a constant, $k \neq -1$.

(i) Show that g is self-inverse. [2]

(ii) Given that $k > 0$, sketch the curve $y = \mathrm{g}(x)$, stating the equations of any asymptotes and the coordinates of any points where the curve crosses the x- and y-axes. [3]

(iii) State the equation of one line of symmetry of the curve in part **(ii)**, and describe fully a sequence of transformations which would transform the curve $y = \dfrac{1}{x}$ onto this curve. [4]